

VoDeMe

Vocabulary Learning Platform

PROJECT TYPE Vocabulary Learning Platform	PLATFORM Responsive full-stack web application
AUDIENCE Grade 10 vocabulary learners and educators	PRIMARY FOCUS Gamified vocabulary development

Overview

A full-stack vocabulary learning platform for Grade 10 students, combining video lessons, quizzes, three themed learning games, and real-time progress tracking. VoDeMe turns synonyms, collocations, roots, prefixes, and suffixes into a structured, gamified experience with secure Google OAuth and database-backed learner progress.

Core Features

- Three multi-level vocabulary games: SynoHit, HopRight, and Word Study Journal
- Video lessons and gated quizzes for synonyms, collocations, and word formation
- Daily vocabulary cards with definitions, examples, synonyms, and antonyms
- Persistent points, lesson completion, quiz results, game progress, and leaderboard ranking
- Google OAuth plus secure email/password authentication with database-backed sessions
- Learner profiles with image uploads and account management
- Post-lesson Learning Wall reflections with community likes
- Responsive, animated learning interface optimized for desktop and mobile

Experience and Design

- SynoHit, HopRight, and Word Study Journal turn three vocabulary domains into themed games.
- Video lessons introduce synonyms, collocations, roots, prefixes, and suffixes before assessment.
- Word of the Day cards combine meaning, usage, synonyms, antonyms, and pronunciation context.
- Points, lesson completion, quiz results, level status, and leaderboards make progress visible.

Architecture and Data

- Next.js App Router and React provide the public site, protected dashboard, lessons, and games.
- Prisma connects server actions and route handlers to Neon PostgreSQL.
- Relational models store accounts, sessions, lessons, quizzes, game progress, and Learning Wall activity.
- UploadThing handles authenticated profile images, while Recharts presents learner analytics.

Security and Privacy

- Google OAuth and Argon2id credentials support secure sign-in options.
- Database-backed sessions use secure httpOnly cookies.
- Short-lived OAuth state records protect Google sign-in against CSRF attacks.
- Zod validation, rate limiting, audit logs, and per-user relations protect application data.

Technology Stack

Next.js 16	Application framework and UI
React 19	Application framework and UI
TypeScript	Application language
Tailwind CSS 4	Interface styling and design system
PostgreSQL (Neon)	Authentication, database, or cloud data
Prisma ORM	Database access and schema management
Google OAuth	Authentication and session security
UploadThing	Media storage and delivery

Performance and Reliability

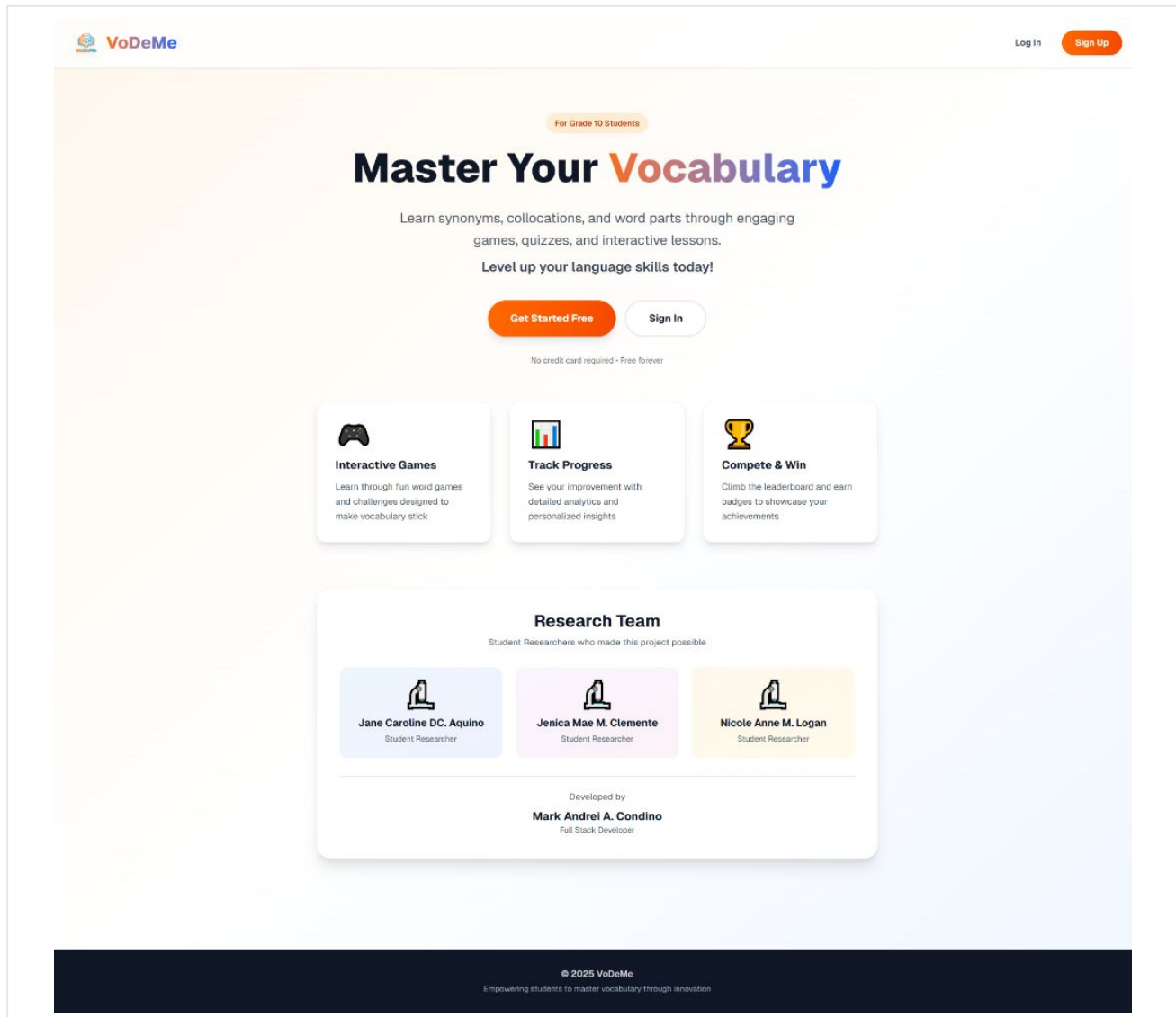
- Server components reduce client-side work for protected dashboard data.
- Thirty-second refresh cycles update stats and rankings without full-page reloads.
- Lesson and game modules load only when learners enter those activities.
- Responsive Tailwind layouts keep the interface usable across desktop and mobile screens.

Platform Requirements

- Current Chrome, Edge, Firefox, or Safari browser with JavaScript enabled.
- Stable internet connection for lesson media, OAuth, and progress synchronization.
- Popup or redirect support for Google account selection.
- A 1280 x 720 or larger display is recommended for the full desktop dashboard.

PAGE OVERVIEW

Landing Page

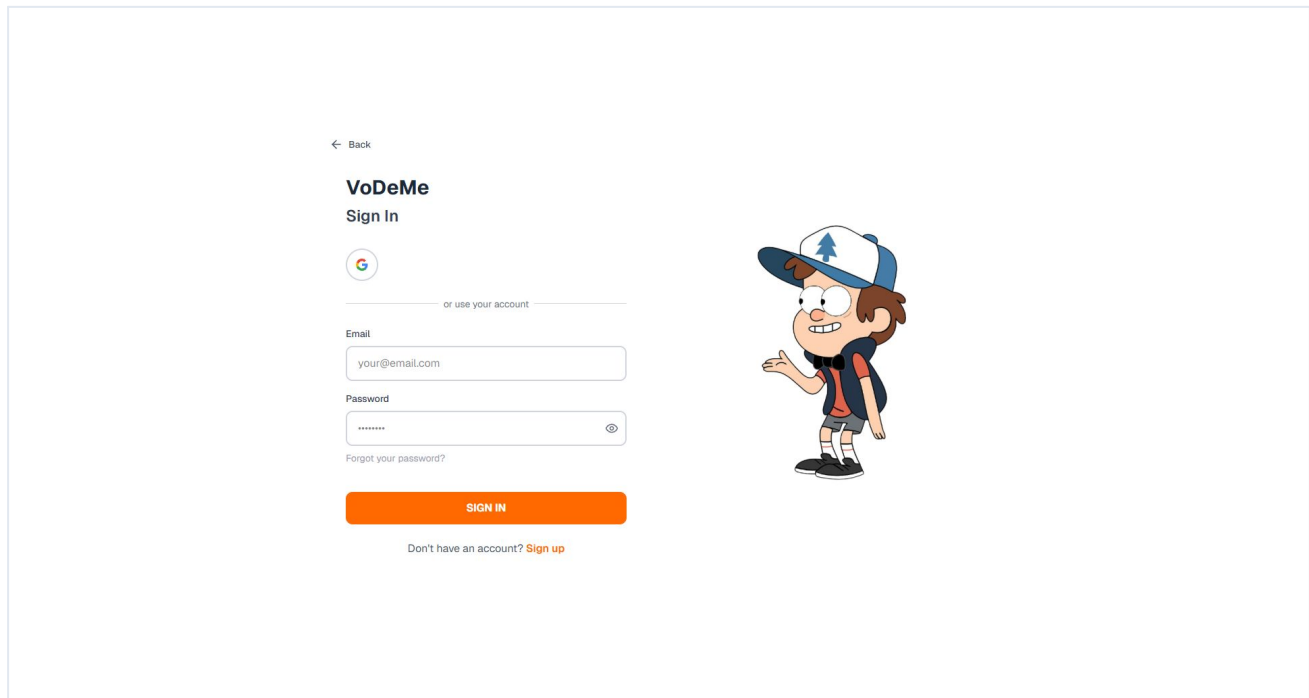


Landing Page

Interface view from VoDeMe.

PAGE OVERVIEW

Sign-In & Google OAuth

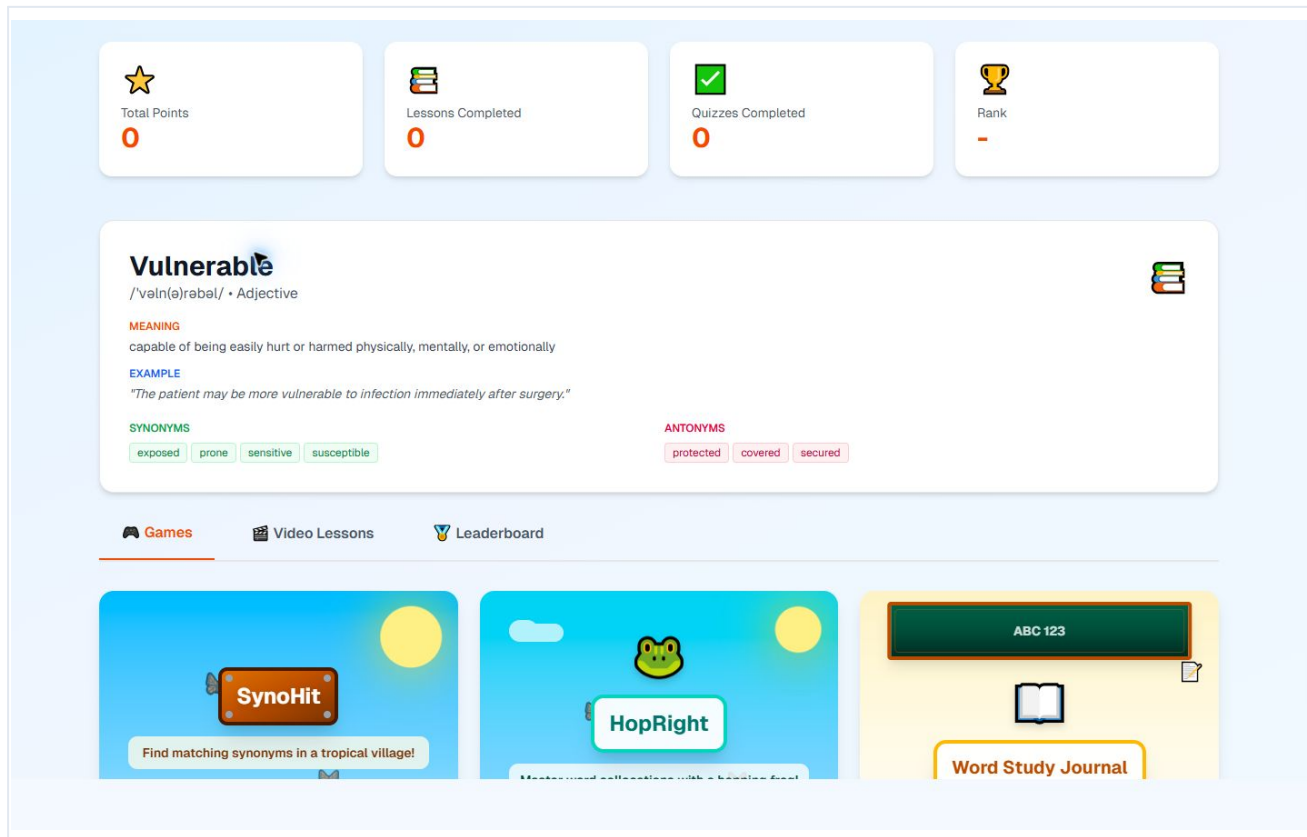


Sign-In & Google OAuth

Interface view from VoDeMe.

PAGE OVERVIEW

Learner Dashboard & Game Library



Learner Dashboard & Game Library

Interface view from VoDeMe.

PAGE OVERVIEW

Video Lessons

Vulnerable

/ˈvʌl(ə)rəbəl/ • Adjective

MEANING
capable of being easily hurt or harmed physically, mentally, or emotionally

EXAMPLE
"The patient may be more vulnerable to infection immediately after surgery."

SYNONYMS
exposed prone sensitive susceptible

ANTONYMS
protected covered secured

[Games](#) [Video Lessons](#) [Leaderboard](#)

Synonyms

Learn how words with similar meanings can be used effectively in different contexts. This lesson demonstrates the importance of choosing the right synonym to convey precise meaning and enhance your vocabulary.

🕒 3:00 [Start](#)

Collocations

Explore common word pairings and their types. This lesson shows how collocations work together naturally in English, helping you sound more fluent and native-like in your language use.

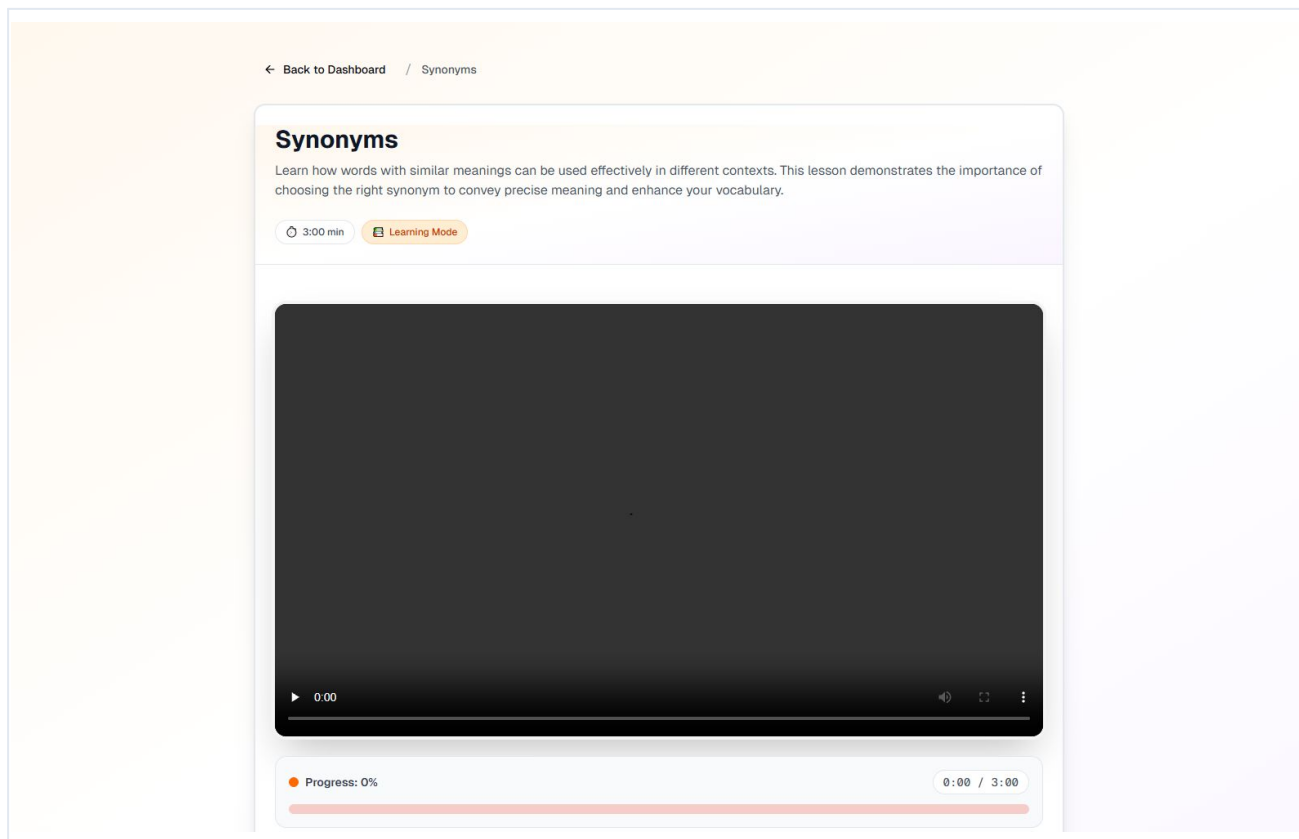
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Video Lessons

Interface view from VoDeMe.

PAGE OVERVIEW

Interactive Lesson Player

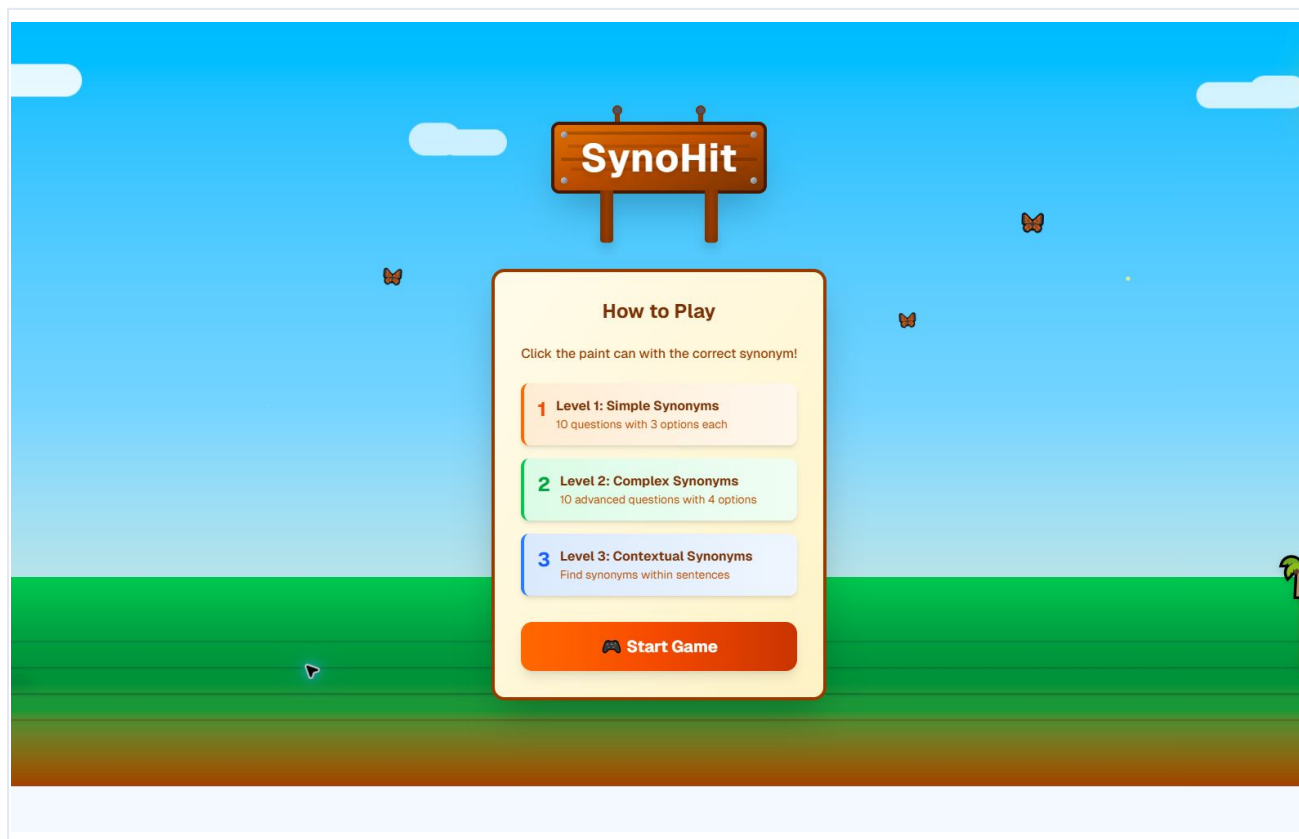


Interactive Lesson Player

Interface view from VoDeMe.

PAGE OVERVIEW

SynoHit - How to Play

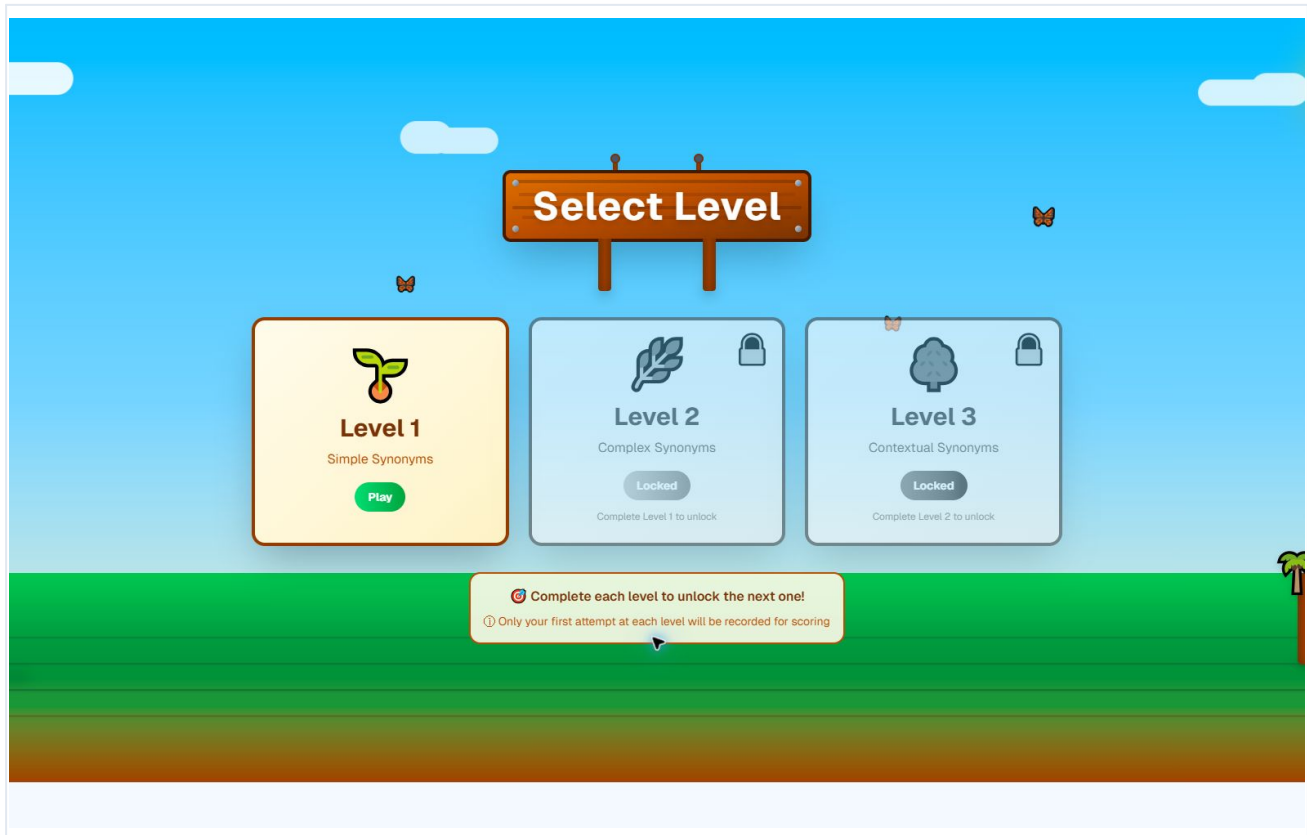


SynoHit - How to Play

Interface view from VoDeMe.

PAGE OVERVIEW

SynoHit - Level Progression

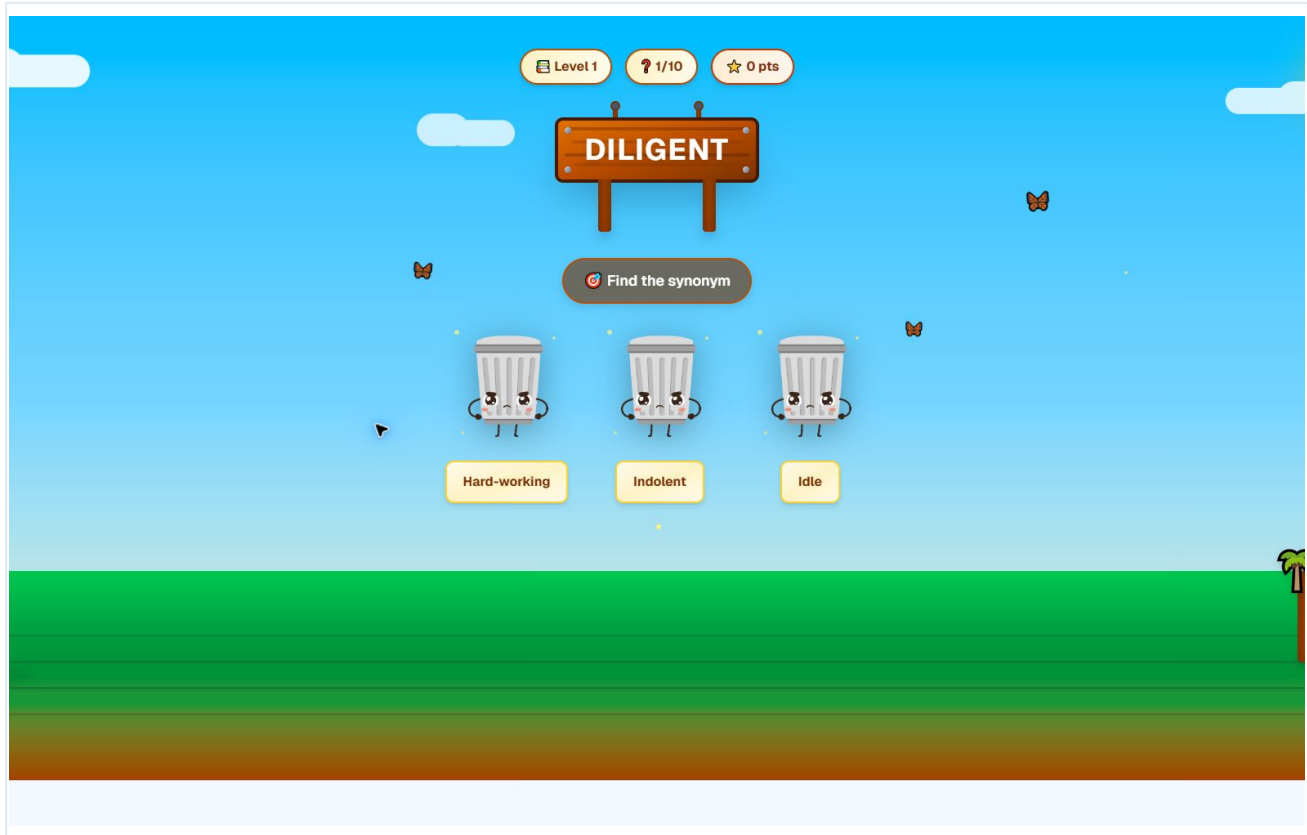


SynoHit - Level Progression

Interface view from VoDeMe.

PAGE OVERVIEW

SynoHit - Live Challenge



SynoHit - Live Challenge

Interface view from VoDeMe.